BBC Sport Text Statistics Analysis

Methodology and data limitations

The supplied BBC Sport files are a preprocessed term-document count matrix, not original article text. Word counts and term statistics are measured from the matrix. Sentence, punctuation, contact details, and original character counts cannot be recovered. Any reconstructed text is synthetic and should not be treated as verbatim source.

Metric	Value
documents	737
categories	5
vocabulary_size	4613
total_term_occurrences	122691
mean_word_frequency	26.597
median_word_frequency	9.0
std_word_frequency	58.181
frequency_percentiles_25_50_75	[5.0, 9.0, 23.0]
average_word_length	5.257
vowel_consonant_ratio	0.601
original_sentence_count	unavailable
average_sentence_length	unavailable

Findings and observations

Frequent terms

The highest-frequency terms are: plai, game, win, england, player, first, against, year, two, time. These terms may reflect shared sports vocabulary rather than category-specific themes.

Category differences

Token totals vary by category: athletics: 15083, cricket: 24328, football: 42183, rugby: 26868, tennis: 14229. Compare vocabulary richness across categories with caution because document lengths vary.

Character-level patterns

The most frequent processed characters are 'e', 'a', 'i', 'r', 't', 'n', 'o', 'l'. Character counts are derived from processed vocabulary, not original article text.

Zipf and anomalies

The log-log frequency curve shows whether a small number of words dominate the corpus. Unusually frequent terms can arise from reporting style or from preprocessing artifacts.

Pattern extraction

Original TXT files analyzed: 0. The matrix does not retain regex-matchable contact details, dates, or URLs.

Interpretation and recommendations

Word frequency follows a heavy-tailed pattern; compare the rank-frequency chart with a Zipf-like reference rather than assuming an exact power law. Frequent terms can reflect general sports reporting and should not automatically be interpreted as distinctive features.

Vocabulary richness is reported as unique terms divided by observed token counts. This metric is sensitive to document length and to the preprocessing used to create the source vocabulary.

Regex challenges and limitations

The original punctuation and context were discarded in the supplied matrix, making faithful detection of email addresses, phone numbers, dates, URLs, and sentences impossible. The program therefore runs regex only on separately provided original TXT files and marks unavailable values explicitly when these are absent. Phone normalization assumes North American ten-digit formats.

Further work

Supply the original BBC articles as TXT files for exact character and sentence statistics, meaningful regex extraction, and reliable n-grams. Review extracted patterns manually to evaluate false positives.